



Calculating theoretical probabilities from Venn diagrams (two events)

1 There are two trials. In Trial 1, a fair spinner with {A, B} is spun twice. In Trial 2, a fair spinner with {B, C} is spun twice. Which statement is true? Tick **1** correct answer

- ☐ The most likely outcome is {B,B}.
- ☒ B is the most likely letter to appear in the trials.
- ☐ C is the least likely letter to appear in the trials.

2 If the probability of an event (A) is $\frac{1}{5}$, and the experiment is repeated 25 times, what is the theoretical number of times event (A) will occur? Tick **1** correct answer

- ☐ 5
- ☒ 10
- ☐ 15
- ☐ 20
- ☐ 25

3 A random character is selected from the word MISSISSIPPI, which of the following statements are true? Tick **2** correct answers

- ☐ The letter selected would most likely be a vowel
- ☒ The letter selected would most likely be a consonant
- ☐ The letter selected would most likely be an 'S'
- ☒ The letter selected would least likely be an 'M'

4 If the probability of an event (A) is $\frac{1}{3}$, and the experiment is repeated 40 times, what is the theoretical number of times event (A) will occur? Tick **1** correct answer

- ☐ 1
- ☐ 8
- ☒ 5
- ☐ 18



5 If you roll a fair six sided die twice and sum the results, what is the most likely sum below? Tick **1** correct answer

☒ 7

☐ 6

☐ 8

6 If you roll a fair six sided die twice and sum the results, what is the most likely sum below? Tick **1** correct answer

☐ an even sum

☐ an odd sum

☒ An even or an odd sum are equally likely

